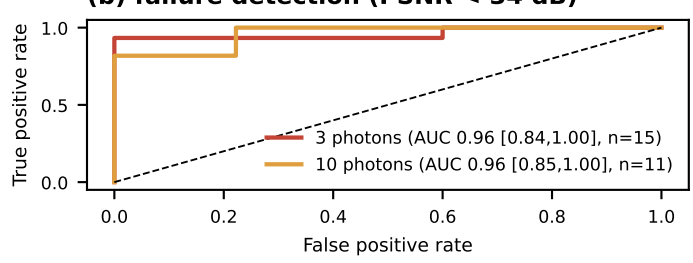
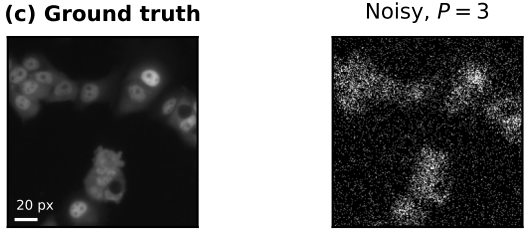
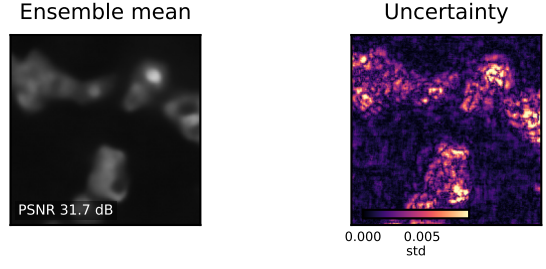
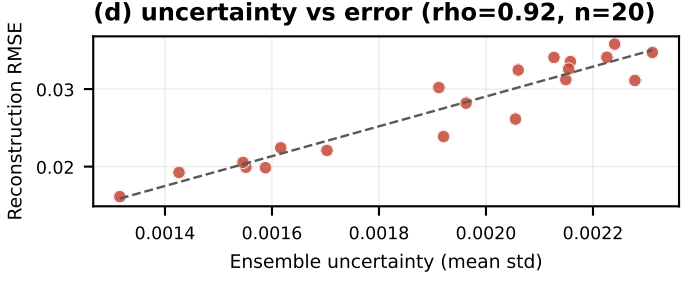
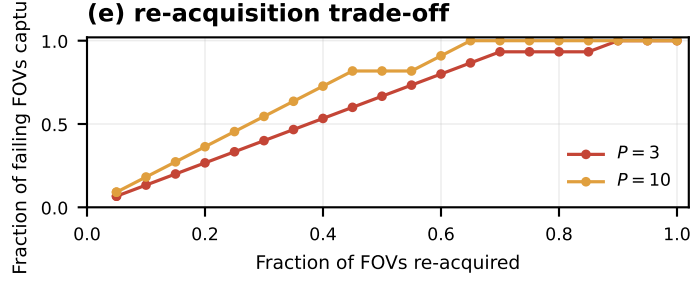

(a) cross-dataset gain at $P = 30$
U-Net PSNR gain (dB)
+1.36
+0.52
+1.30
+0.16
BBBC014
BBBC015
BBBC016
BBBC018


(b) failure detection (PSNR < 34 dB)
True positive rate
False positive rate
3 photons (AUC 0.96 [0.84,1.00], n=15)
10 photons (AUC 0.96 [0.85,1.00], n=11)


(c) Ground truth
Noisy, $P = 3$
20 px


Ensemble mean
Uncertainty
PSNR 31.7 dB
0.000
0.005
std


(d) uncertainty vs error (rho=0.92, n=20)
Reconstruction RMSE
Ensemble uncertainty (mean std)


(e) re-acquisition trade-off
Fraction of failing FOVs captured
Fraction of FOVs re-acquired
$P = 3$
$P = 10$